

Name: _____

The Ripple Effect — The Work of Water Across New York State Surfaces **Student Directions**

Focus Questions: While performing this investigation, use the following focus questions to guide your work.

1. How is Earth’s surface changed through erosional and depositional processes?
2. How can measurements of permeability of different materials be used as evidence for the predicted amount of infiltration, surface runoff, and retention of water?
3. How does water, in its different forms, chemically and mechanically cause changes in New York State rocks?
4. How can water cycle processes change Earth’s surface and create feedbacks that cause changes to Earth systems?
5. How can models be used to illustrate water’s effects on surface processes that operate at different spatial and temporal scales?
6. What effects do the properties of water have on Earth materials and surface processes?
7. How does water affect the melting points of rocks?

Purpose of Investigation: Earth is often called the water planet because of the abundance of liquid water on Earth’s surface. Water’s unique physical and chemical properties are central to its effects on Earth materials and surface processes. In this investigation, you will choose either **Investigation Option A** or **Investigation Option B** in order to plan and conduct an investigation using models to collect data to examine large-scale Earth surface processes on a smaller temporal and spatial scale. The results of this investigation will provide evidence for the effect of water on Earth materials and surface processes. All students will also complete Investigation C.

In **Session 1: Part 1** you will collaboratively develop a written investigation plan, based on given criteria and constraints, to produce data to be used as evidence for the effect of water on Earth’s materials and surface processes. You will record this investigation plan in ***Student Answer Packet 1***.

Your tasks in **Session 1: Part 1** will be to:

- plan an **Investigation Option, A or B**
- identify the phenomenon (variable) to be investigated for your option (a connection between the properties of water and its effects on Earth materials and surface processes).
- make a prediction about how the identified phenomenon is affected by water
- develop a written investigation plan that includes a title, purpose, list of materials, and a step-by-step procedure that describes the data that will be collected and how the data is relevant for determining the effect of water on Earth materials and surface processes, based on given criteria and constraints

Your tasks in **Session 1: Part 2** will be to:

- continue to collaborate in pairs to conduct your chosen investigation
- gather measureable (quantitative) data to be used as evidence for the effects of water on Earth materials in ***Student Answer Packet 1***
- communicate your data to the rest of the class for analysis
- perform **Investigation C**
- record the data from the other **Investigation Option (A or B)** in ***Student Answer Packet 1***

Your task in **Session 2:** will be to apply what you learned in **Session 1** to independently answer the investigation questions found in ***Student Answer Packet 2***.

Investigation Option A: Stream Table

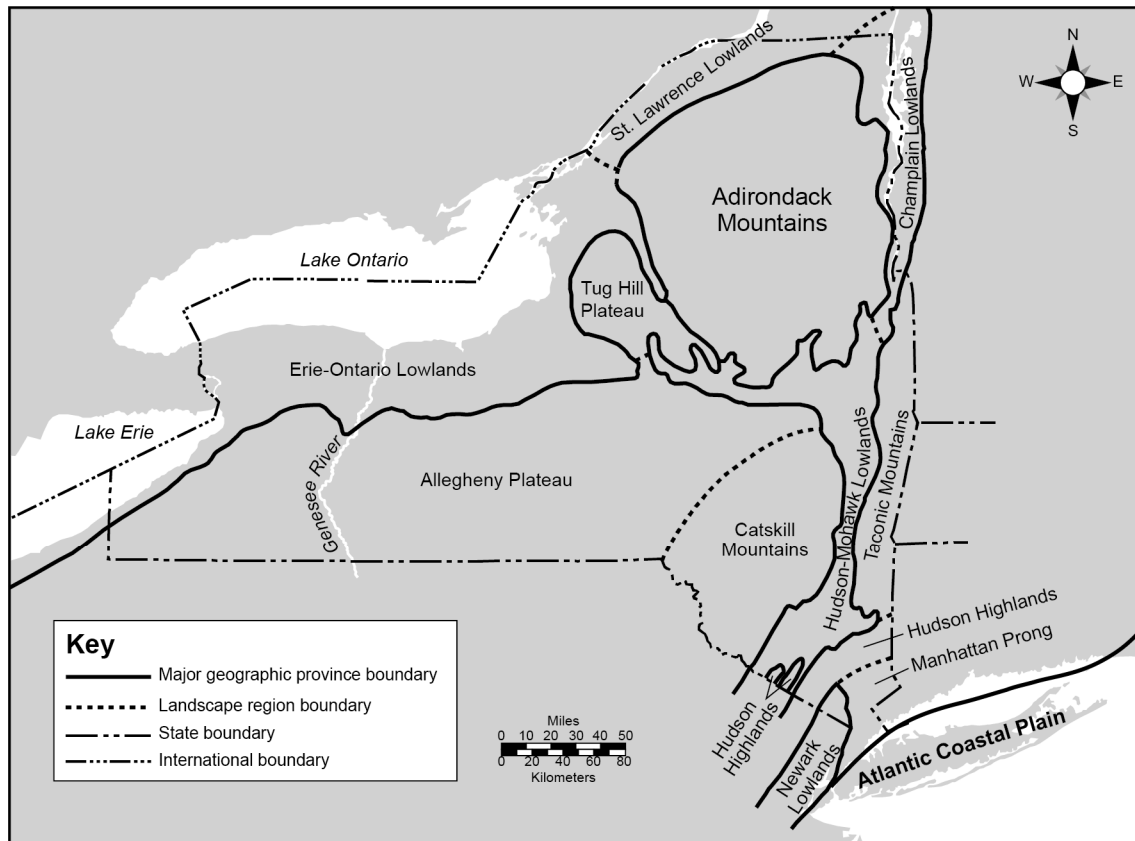
The Genesee River is an important waterway in New York State. It flows from south to north across the Allegheny Plateau and Erie-Ontario Lowlands, depositing sediment into Lake Ontario near Rochester, New York. Streambank erosion within the Genesee River Basin creates problems by placing a huge amount of sediment and silt into the stream, creating a brown and muddy appearance, which negatively impacts fish and invertebrate populations throughout the basin.

Photograph of Genesee River



During periods of heavy rain or snowmelt, this river can experience flooding, which can inhibit infiltration into groundwater systems.

Landscape Regions of New York State

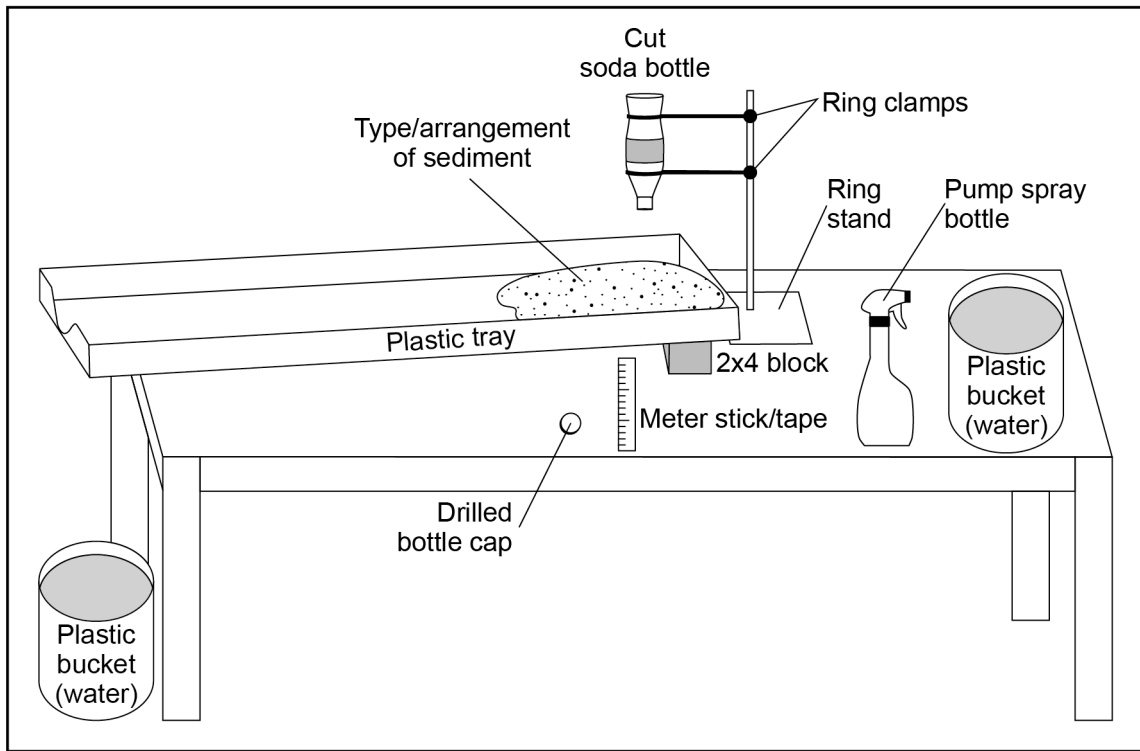


The goal of **Investigation Option A** is to answer **Focus Questions one, four, five, and six**:

1. How is Earth's surface changed through erosional and depositional processes?
4. How can water cycle processes change Earth's surface and create feedbacks that cause changes to Earth's systems?
5. How can models be used to illustrate water's effects on surface processes that operate at different spatial and temporal scales?
6. What effects do the properties of water have on Earth materials and surface processes?

You and your partner, choosing from the materials available, will plan and conduct an investigation to collect quantitative data to be used as evidence for the effect of water on different Earth materials, which contribute to erosion, deposition, or changes to Earth's surface.

Model of Stream Table Setup



List of Possible Materials

• stream table or plastic (under the bed) storage tote 41 qt. with a 5/8-in. hole drilled in the center of one short side 1/2 in. from the bottom • washcloth (to simulate a vegetated surface) • acetate sheet (to simulate an urban surface) • meter stick/tape	• 2-L soda bottle with the bottom cut off and a 1/4-in. hole drilled in the cap or 16-oz plastic cup with evenly distributed holes in bottom • fine sand – wet • coarse sand – wet • gravel (pebble-sized particles) – wet • figurines* (like Monopoly houses) • ring stand and ring clamp	• 2x4 blocks of wood • collection bucket (2-2.5 gal. bucket/paint bucket) • water • spray bottle (for wetting sediment) • stopwatch • acetate sheets – cut lengthwise (to be used to divide stream table)
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* For the purpose of producing quantitative data

Criteria/constraints for completing Investigation Option A:

- can use one size of sediment alone or a mix of different-sized sediments (e.g. 75% fine sand and 25% gravel or 50% medium sand and 50% gravel)
- use enough sediment to create a landscape that allows for quantitative evidence of erosion and deposition (e.g. by measuring the distance particles/figurines move)

Investigation Option B: Rain Garden

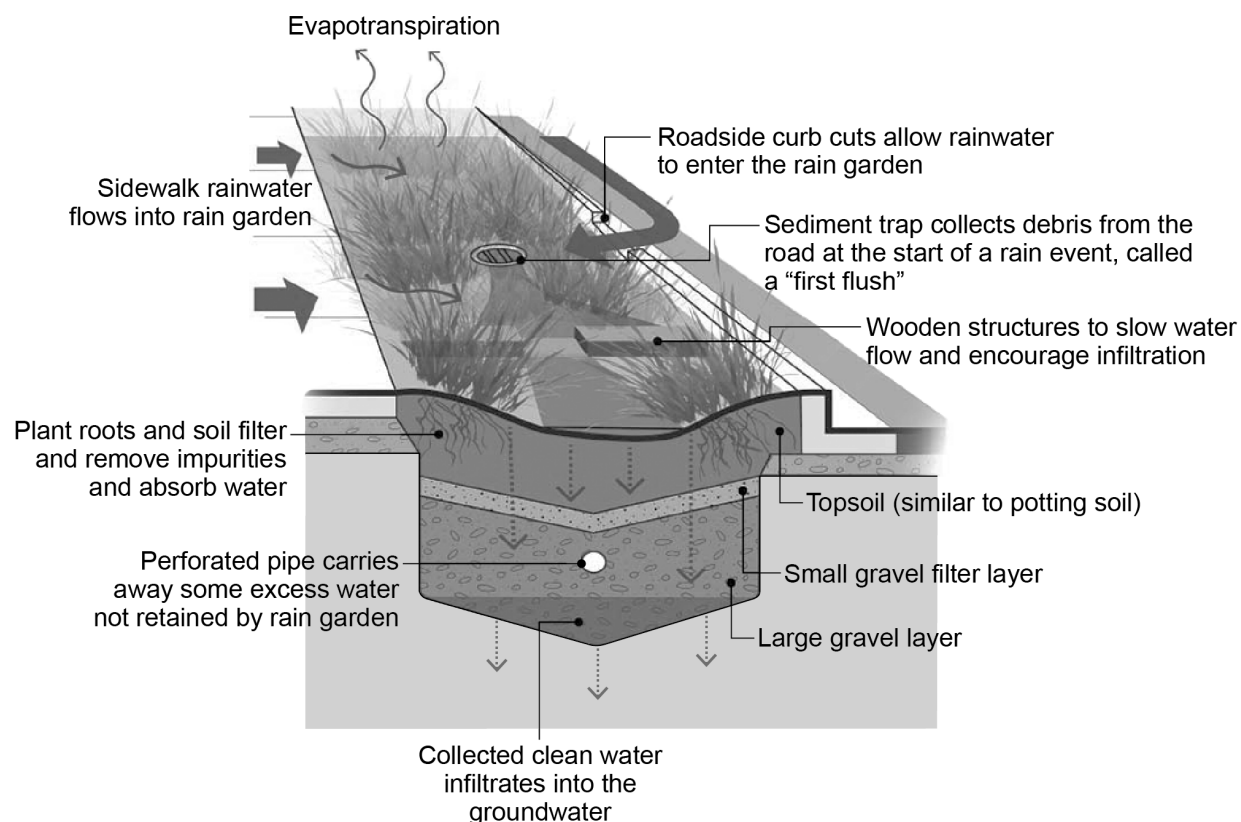
Rain gardens in New York City, located in the Manhattan Prong landscape region, are designed to capture and absorb stormwater runoff from impermeable surfaces such as rooftops, streets, and sidewalks. Rain gardens are a key component of New York City's green infrastructure plan, using vegetation, soil, and other materials to capture and manage stormwater runoff and prevent flooding.

Rain gardens are green infrastructures that are designed and constructed to capture stormwater runoff in urban areas to promote infiltration and retention (storage). Runoff in areas without rain gardens may enter the combined sewer system (storm sewers and sanitary sewers), which can exceed capacity and lead to overflows, causing contamination to nearby waterways. Effective rain gardens absorb rain water and reduce polluted stormwater runoff and runoff entering the combined sewer system.

New York City Rain Garden



Green Infrastructure Rain Garden Model



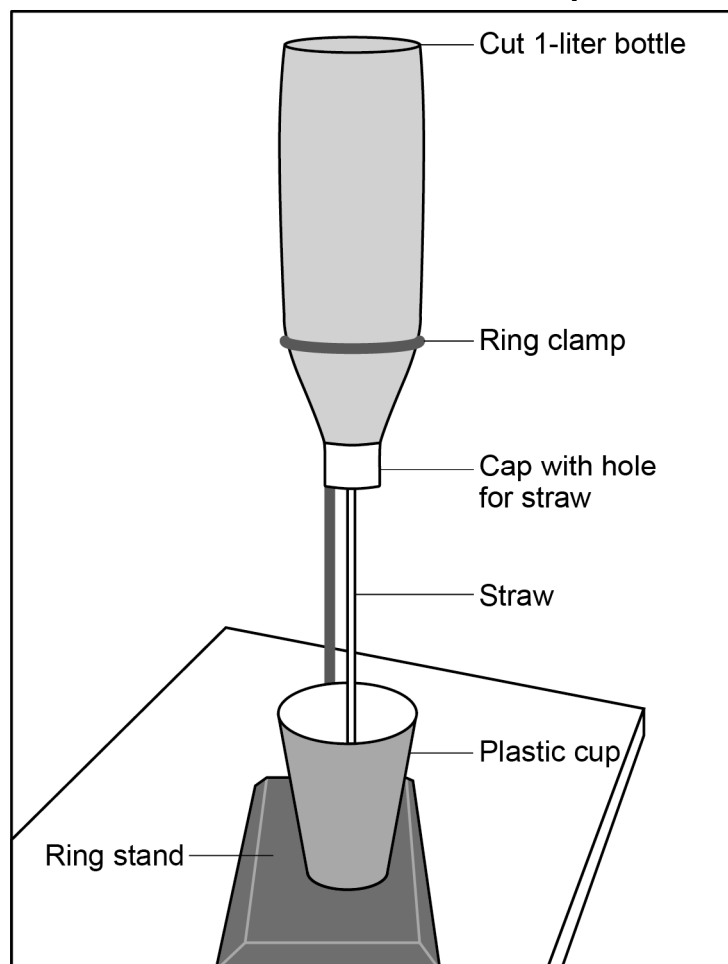
The goal of **Investigation Option B** is to answer **Focus Questions two, four, and five**:

2. How can measurements of permeability of different materials be used as evidence for the predicted amount of infiltration, surface runoff, and retention of water?
4. How can water cycle processes change Earth's surface and create feedbacks that cause changes to Earth systems?
5. How can models be used to illustrate water's effects on surface processes that operate at different spatial and temporal scales?

You and your partner, choosing from the materials available, will create a model of a "rain garden" to collect data and to test the effect of different materials on permeability, infiltration, runoff, and water retention. This data will be used as evidence to determine which materials should be used in the most effective urban rain garden designs. Effective designs should retain surface runoff while controlling infiltration into the ground, therefore reducing the amount of water entering the combined sewer system and minimizing contamination of nearby waterways.

The chosen materials to be placed in the 1-liter bottle are not shown in the model.

Model of Rain Garden Setup



List of Possible Materials

• 1-L clear soda bottle with the cap drilled to the diameter of a straw • 16-oz. bottle of any type – cap predrilled with 1/8-in. holes (to simulate a rain shower) • straw • ruler	• coffee filter • cheese cloth • cotton balls • stopwatch • ring stand and clamp • food coloring (blue or red) • 100-ml graduated cylinder • spoons	• paper towels • beakers • gravel (pebble-sized particles) – wet • fine sand – wet • coarse sand – wet • potting soil – wet
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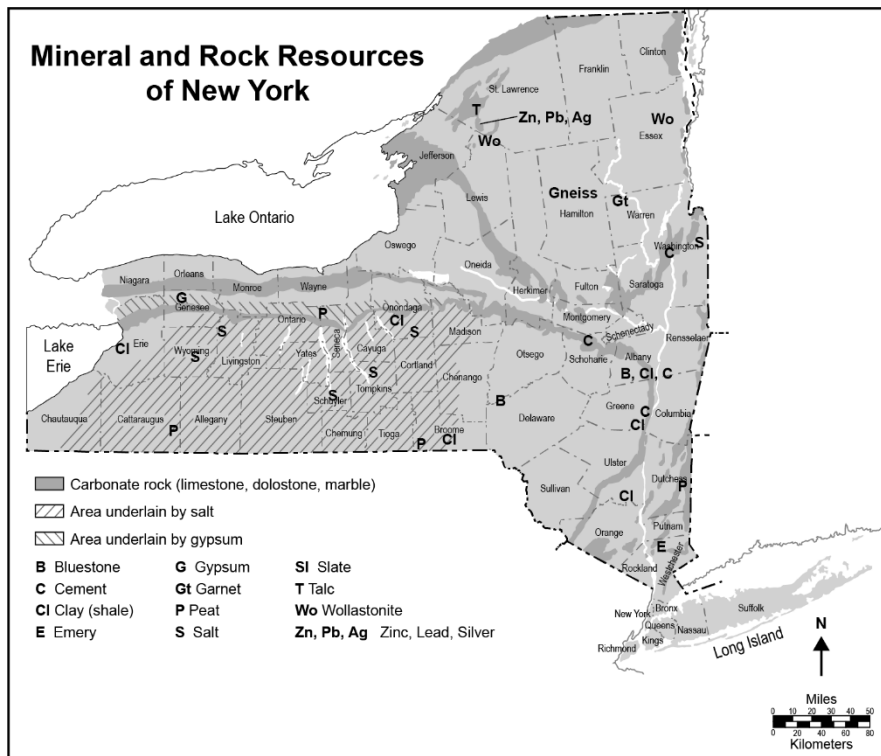
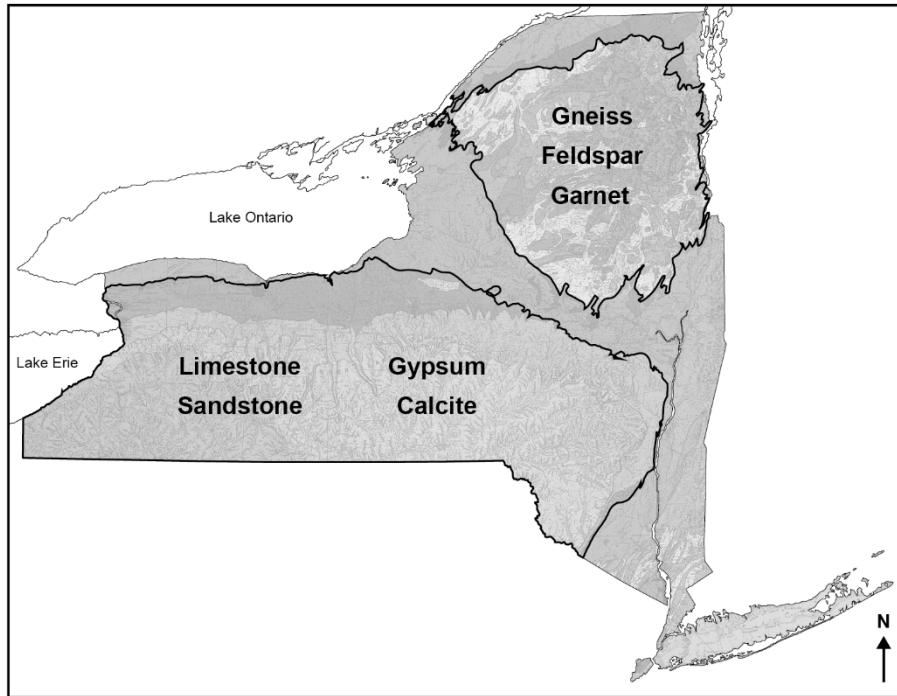
Criteria/constraints for completing Investigation Option B:

- should not exceed 400 to 500 ml of sediment in the bottle (one full 16-oz. plastic cup)
- all sediments should be wet before conducting the investigation
- colored water can be used
- only enough food coloring drops to achieve an observable color change in the water should be used
- the top of the straw must not extend into the bottle
- no sediment should be transported into the bottom cup as the focus of the investigation is flow of water, not sediment
- the rain simulator may need to be tilted or vented to achieve a continuous flow of water into the rain garden

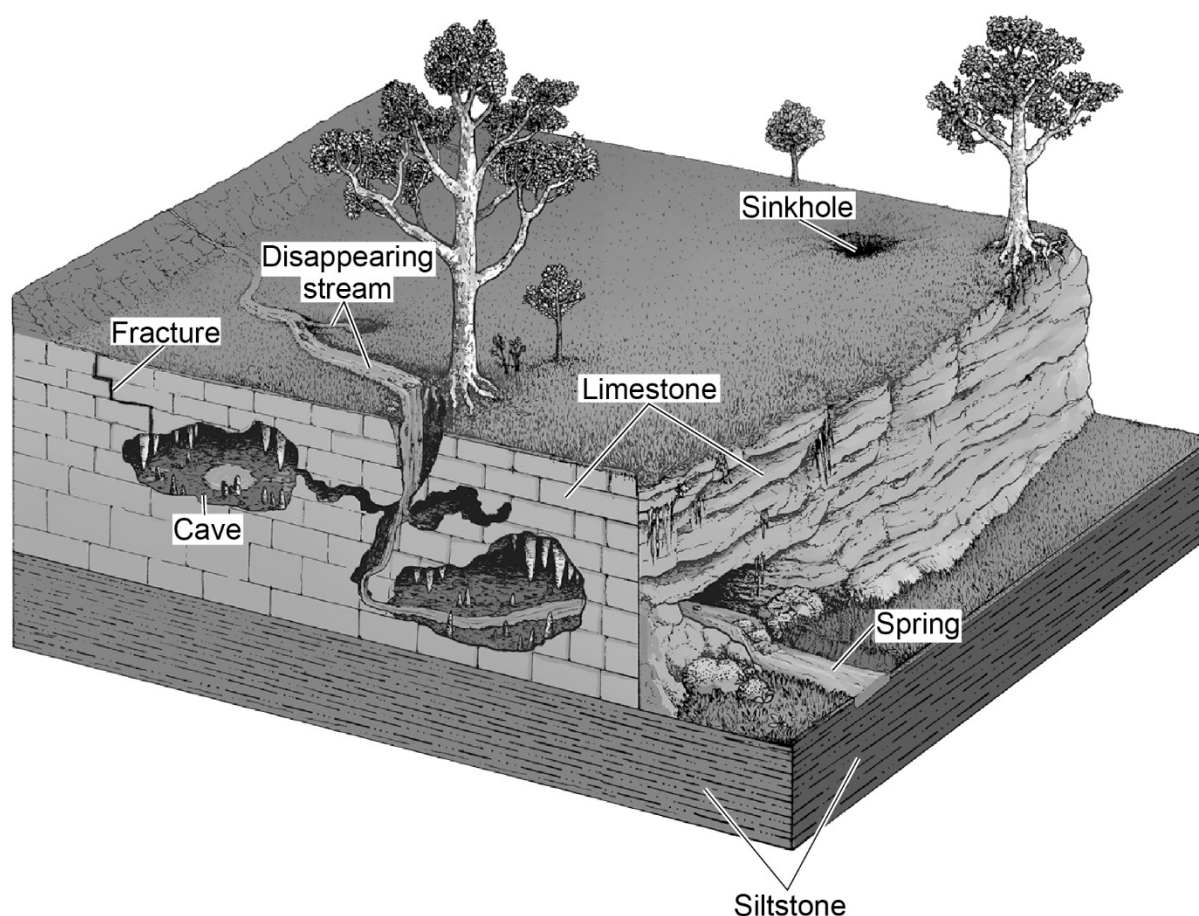
Investigation C: Weathering of Earth Materials

Weathering of rocks, which occurs at different spatial and temporal scales, has played a significant role in shaping the surface of New York State landscape regions. Different rock types found in the Allegheny Plateau and the Adirondack Mountains experience different rates and types of weathering.

Approximate Locations of Some Rocks and Minerals in New York State



The unique properties of water in its different forms are central to water's ability to physically and chemically weather rocks. These properties include water's ability to dissolve substances and water's ability to expand as water freezes, creating pressure within rocks.



The goal of **Investigation C** is to answer **Focus Questions three** and **six**:

3. How does water, in its different forms, chemically and mechanically cause changes in New York State rocks?
6. What effects do the properties of water have on Earth materials and surface processes?

You and your partner will collect data to be used as evidence on the effects of water on rocks and minerals in New York State. The chalk used in this investigation represents limestone formations found across New York State. The tap water represents precipitation and the vinegar represents acid rain conditions that affect Earth materials and surface processes. Additionally, you will describe how these processes contribute to the formation of unique geological features found in different landscape regions.

All students will use the materials for **Investigation C**.

List of Possible Materials

- chalk – two pieces (approximately 2.5 - 3.5 cm in length)
- white vinegar
- 3.5 oz. transparent plastic cups – two
- electronic balance
- masking tape
- tap water
- stopwatch
- goggles
- tweezers

Procedure:

Session 1: Part 1 (Planning the Investigation)

1. Write your name on ***Student Answer Packet 1***.
2. Read and review the information about the **Investigation Options (A and B)**.
3. In ***Student Answer Packet 1***, indicate the letter of the **Investigation Option** that you and your partner are conducting.
- 4a. In ***Student Answer Packet 1***, identify the independent and dependent variables.
- 4b. Make a prediction that specifies what happens to the dependent variable when the independent variable is manipulated.
5. With your partner, plan an investigation to determine the effect of your independent variable on Earth's materials and surface processes. Be sure to control all variables except the one you are investigating and perform three trials for each value of the independent variable.

Criteria: Your Investigation Option Plan must include...

- a title
- the purpose of the investigation
- a description of the numerical data, including units, to be collected
- evidence derived from the data
- a description of how the data collected will be relevant to determining the effect of water on Earth materials and surface processes
- a selection of appropriate tools/materials to collect, record, analyze, and evaluate data
- a step-by-step procedure that must be focused on the purpose of the investigation and be specific enough so that other pairs of students can conduct the exact same investigation by reading your procedure
- a labeled model of your chosen investigation setup
- a student-designed data table to record numerical data

Constraints:

- you are limited to the materials provided for each option
- your investigation should be completed in approximately 40 minutes (one class period)
- you **may not** taste any substance

6. Create an investigation plan (a written, step-by-step procedure), based on the criteria and constraints.

Each student needs to record the investigation plan in their own **Student Answer Packet 1**.

7. Create your data table in **Student Answer Packet 1**. Include a title, dependent variable, independent variable, three trials, averaged data, and any observations.

Session 1: Part 2 (Conducting the Investigation)

8. Gather all the materials necessary to perform your planned investigation. Set up your investigation.
9. Complete and label the model in **Student Answer Packet 1** to show your setup with investigation variables. For each design change, a new model setup must be completed. Attach revised models to the back of **Student Answer Packet 1**.
10. Working with your partner, conduct your planned investigation. Make sure to record all your numerical data for each trial in the data table for your **Investigation Option**. Also, record any relevant observations while conducting your investigation.
11. Clean up your investigation materials and return all materials to their original location.
12. Evaluate your collected data with your partner to make a claim using numerical data in **Student Answer Packet 1** about whether the data supports or refutes your prediction. (You will only need to make a claim about the **Investigation Option** you and your partner performed.)
13. Share your data table with the rest of the class, as directed by your teacher. Record the data from the other **Investigation Option** displayed by your teacher into the data table in **Student Answer Packet 1**.
14. Indicate the students' names that you obtained your data from in the data table.

15. Gather all the materials necessary to perform **Investigation C**.
16. Predict the type of weathering that will most likely occur in this investigation. Identify one piece of observational evidence that would support this prediction.
17. Make a prediction about which type of liquid will cause the greatest amount of weathering to the chalk. Give a reason for your answer.
18. Use appropriate safety equipment and follow all safety procedures as determined by your teacher when working with the vinegar and chalk.
19. Working with your partner, you will collect data as evidence for how different liquids (tap water and vinegar) affect the weathering of an Earth material (calcium carbonate chalk).
20. Label one cup “vinegar” and one cup “tap water” using masking tape. The “vinegar” represents acid rain and the “tap water” represents normal precipitation.
21. Mass each piece of chalk and record this value in the data table, *Effect of Type of Liquid on Amount of Weathering*, in **Student Answer Packet 1** in the “Initial” column.
22. Place one piece of chalk in the bottom of each cup.
23. Cover each piece of chalk with equal amounts of the labeled liquids.
24. Begin timing and record your observations in the data table about changes to the chalk or liquid.
25. After 5 minutes, carefully remove each piece of chalk from the labeled liquid.
26. Mass each piece of chalk and record this value in the data table on **Student Answer Packet 1** in the “Final After 5 minutes” column.
27. Calculate the change in mass for each piece of chalk and record this value in the data table in **Student Answer Packet 1** in the “Change” column.
28. Evaluate your collected data with your partner to make a claim at the bottom of your data table in **Student Answer Packet 1** about whether the data supports or refutes your prediction.
29. Return your **Student Directions** and **Student Answer Packet 1** to your teacher.

Session 2

Your task in **Session 2** is to apply what you learned in **Session 1** to independently answer the investigation questions found in **Student Answer Packet 2**.